

Study 1

Pilot

Random split of overlapping sequences;
30 optimiser steps;
baselines with all labels

FedGTCL ahead;
baselines collapse

Study 2

Development and held-out tests

Configurations tuned on an inner split;
CICAPT-IIoT2024 and X-IIoTID held out

No method learns on either held-out set

Study 3

Window-disjoint, equal budget (sealed)

Chronological regions;
1,000 steps; equal labels;
FedProx and pseudo-labelling added

Not better than FedProx;
its own components hurt

Study 4

Revised variant, external test (sealed)

Fixed-rule selection;
re-test on new partitions;
test on UNSW-NB15

Wins internally,
loses externally

Study 5

Controlled factors (sealed)

Split x budget x baseline labels
on four datasets

Budget flips the ranking;
leakage raises IIoT results

Study 6

Budget curve and re-tuning (sealed)

Up to 3,000 steps with checkpoints;
learning rate re-tuned on an inner split

FedGTCL ahead at 200-400 steps,
behind from 1,000

Study 7

Per-record classifiers (sealed)

Random vs temporal record split; LR and MLP;
three datasets

Random split: MCC +0.15,
accuracy +0.03